Reg. No. | | | | | | | | | | [13AE63/12AE63]

SIXTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, JUNE 2016

PROPULSION-II

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	Derive the rocket equation.	**09**
	b.	Solve the rocket equation to obtain an expression for velocity increment.	**05**
2.		Explain the following performance parameters of rocket engines.	
	a.	Specific impulse	**03**
	b.	Vehicle mass ratio	**02**
	c.	Propellant mass fraction	**02**
	d.	Effective exhaust velocity	**02**
	e.	Thrust coefficient	**03**
	f.	Characteristic velocity	**02**
3.	**a.**	A mixture of ideal gases consists of 3kg Nitrogen and 5kg carbon dioxide at a pressure of 300kPa and a temperature of 20^0C. Find the mole fraction of each constituent and the equivalent molecular weight of the mixture.	**04**
	b.	What is	
		i. Absolute enthalpy	**02**
		ii. Enthalpy of formation	**03**
		iii. Upper heating value	**01**
	c.	Explain adiabatic flame temperature.	**04**
4.	**a.**	What are the primary functions of the injector?	**04**
	b.	Write a note on divergence losses.	**04**
	c.	Why $LH_2 - LOX$ rockets run fuel-rich?	**06**
5.	**a.**	List the desirable properties of solid propellants.	**06**
	b.	Define the term linear burning rate. What is its relation with chamber pressure?	**04**
	c.	A solid rocket motor has an end-burning grain of cross-sectional area $0.4m^2$. The density of the propellant is $1500kg/m^3$ and has linear regression rate 5mm/s. If the specific impulse of the propulsion system is 200s, calculate the thrust produced by the motor.	**04**

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6.	**a.**	Describe the cooling methods used in liquid propellant rockets.	**09**
	b.	A liquid rocket engine with oxidizer to fuel ratio of 5:1 produces a thrust of 1MN. Initial mass of the rocket is 100,000 kg and its mass at burn out is 10,000kg. The characteristic velocity is 2386 m/s and thrust coefficient is 1.4. What is the mass flow rate of fuel?	**05**
7.	**a.**	Sketch neatly the staged combustion cycle and give its salient features.	**09**
	b.	Write short note on hybrid rockets.	**05**
8.	**a.**	Explain the significance of multi-staging in rocket propulsion.	**08**
	b.	A rocket with a total lift off mass 10,000 kg moves vertically upward from rest under a constant gravitational acceleration of 9.81 m/s^2. The vehicle mass ratio is 0.16 and the propellant burns at a constant rate of 1200 kg/s. If the specific impulse of the rocket engine is 240s, calculate the velocity at burnout.	**06**